環保染色方案
ECO Dyeing Solution



- **實現無鹽無鹼染色**
  No Salt, No Alkali
- **染色殘液幾乎沒有顏色**
  Almost No Color in Effluent
- **邁向 2050 淨零排放的有效方案**
  Effective & Efficient Solution Toward 2050 Net Zero Emission

- 適用於纖維素織物
- Evertex CA-01 纖維反應型陽離子化助劑
- Everclear / Everclear s/p 反應性染料
- 降低染料用量 40%
- 免去排放可溶性鹽類達 640 公斤／噸織物
- 節省水資源耗用與汙水排放 40%
- 減少時間與電力消耗 35%
- 節省蒸氣熱能 85%，降低碳排放 72%
- 明顯降低廢水 COD 達 80%
- 幾乎無有色度廢水，水資源可高度回用
- 均染性、再現性、濕牢度優異
- 沒有合成纖維廢棄物與微塑膠問題

- For cellulosic substrate dyeing
- Evertex CA-01 fiber reactive type cationic reagent
- Everclear / Everclear s/p reactive dyestuff
- Decrease 40% dye dosage
- Prevent using salt, soda ash 640 kg/ton-fabric
- Save valuable water and reduce effluent 40%
- Reduce process time and electricity consumption 35%
- Save 85% steam energy, reduce 72% carbon emissions
- Reduce effluent COD up to 80%
- Almost no color effluent, water easily be recycled
- Unique levelness, reproducibility and wet fastness
- Free from plastic waste and microplastic problem

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Warnings: *The information and recommendations presented here were completed with the utmost care, but can't be extended to cover every possible case. They are intended to serve as non-binding guidelines and must be adapted to the prevailing condition. The product whether used singly or in combination with other substances, should be carried out in lab test prior to bulk production. The user needs to assume all risk and liability no matter whatsoever results are.*

bluesign SYSTEM PARTNER

July 2025

Everclear® Dyeing Solution

無鹽無鹼反應性染料染色
Salt and Alkali Free Reactive Dyeing

Effective & Efficient Solution Toward 2050 Net Zero Emission

Everlight Chemical

Everclear® (1/1)N;Depth%		染色樣 Specimens	水洗牢度 Wahsing Fastness ISO 105 C06-C2S			耐汗牢度 (酸) Acid Perspiration ISO 105 E04			耐汗牢度 (鹼) Alkali Perspiration ISO 105 E04			耐光牢度 20AFU Lightfastness AATCC TM 16.3-3
			C.C	C.S	N.S	C.C	C.S	N.S	C.C	C.S	N.S	Fading
Yellow G	2.92%		4-5	5	5	4-5	4-5	5	4-5	5	5	4
Yellow R	1.83%		4-5	5	5	4-5	4-5	5	4	4-5	5	4-5
Orange R	1.85%		4-5	4-5	5	4-5	4-5	5	4-5	4-5	5	3
Scarlet G	2.04%		4	4-5	5	4-5	4-5	5	4-5	4-5	5	3-4
Red B	1.81%		3-4	4-5	5	4	4-5	5	4	4-5	5	4
Red 2B	1.89%		4	4-5	5	4	4-5	5	4	4-5	5	3-4
Sky Blue G 1/3N	1.48%		4-5	4-5	5	4-5	4-5	5	4-5	4-5	5	4
Blue G	3.22%		3-4	4-5	5	4-5	4-5	5	4-5	4-5	5	3-4
Blue R	3.84%		4-5	4-5	5	4-5	4-5	5	4-5	4-5	5	3
Navy R	1.78%		4	4-5	5	4-5	5	5	4-5	5	5	3
Yellow R s/p	2.00%		4-5	4-5	5	5	4-5	5	5	4-5	5	3-4
Yellow 2R s/p	1.76%		4-5	4-5	5	5	4-5	5	5	4-5	5	4
Amber R s/p	1.10%		4-5	4-5	5	4-5	4-5	4-5	4-5	4-5	4-5	3-4
Red G s/p	1.19%		5	4-5	5	5	4-5	5	5	4-5	5	2-3
Red B s/p	1.92%		5	4-5	5	5	4-5	5	5	4-5	5	3-4
Rose B s/p	2.42%		4-5	4-5	4-5	4-5	4	4-5	4-5	4	4-5	3
Bordeaux R s/p	1.08%		5	4	5	5	4-5	4-5	4-5	4-5	5	2-3
Blue G s/p	2.79%		5	4-5	5	5	4-5	5	5	4-5	5	4-5
Sapphire B s/p 1/3, additional fixed	2.50%		4-5	4-5	4-5	4-5	4-5	4	4-5	4-5	4-5	3-4
Dark Blue G s/p	1.63%		4-5	5	5	4-5	4-5	5	4-5	4-5	5	3

Everclear® (1/1)N;Depth%		染色樣 Specimens	水洗牢度 Wahsing Fastness ISO 105 C06-C2S			耐汗牢度 (酸) Acid Perspiration ISO 105 E04			耐汗牢度 (鹼) Alkali Perspiration ISO 105 E04			耐光牢度 20AFU Lightfastness AATCC TM 16.3-3
			C.C	C.S	N.S	C.C	C.S	N.S	C.C	C.S	N.S	Fading
Deep Orange s/p	1.26%		4-5	5	5	4-5	4-5	5	4-5	4-5	5	4
Deep Red s/p	1.63%		4-5	5	5	4-5	4-5	5	4-5	4-5	5	4
Deep Navy s/p	1.53%		4	5	5	4-5	5	5	4-5	5	5	2-3
Deep Black-01 s/p CIE L: 14.70	4.0%		4-5	4-5	5	4-5	4-5	5	4-5	4	5	3-4
Deep Black-02 s/p CIE L: 14.96	6.0%		4-5	5	5	4-5	5	5	4-5	5	5	3-4
Deep Black-03 s/p CIE L: 15.32	6.0%		4-5	5	5	4-5	5	5	4-5	5	5	3-4

Everclear® 環保染色方案 Everclear® Dyeing Solution

傳統反應性染料染纖維素基材，需耗用大量的中性鹽、鹼劑，中深色染料用量高、固著率低，染後需繁瑣的溫熱皂洗，不僅耗用大量化學品、蒸氣、電力與水資源，還產生大量難以處理的廢水。

Everclear 環保染色方案，有別於傳統反應性染料染色，僅需使用 Evertex CA-01 預先對纖維素基材施與簡單的陽離子化預處理，即可以使用非常簡易的方式進行染色。染色過程無需添加中性鹽與鹼劑，染色後殘液色度值僅幾乎為零，亦僅需簡單之冷水洗清即完成，整個程序不會產生高色度廢水，殘液可輕易回收或回用。

Everclear 環保染色方案，讓反應性染料染色不再是一個惱人的困擾，可免除使用大量的化學品，顯著節省可貴的蒸氣、電力與水資源，提升生產品質、再現性，並穩定濕牢度，大大地提升生產效能。

Traditional dyeing of dark colors with reactive dyes requires high dosage of dyes, salts and alkali. Due to the poor color fixing effect, it is not only difficult to wash-off, but also consumes a lot of water and energy. It also produces large amounts of darker-colored wastewater.

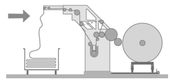
Everclear Dyeing Solution is different from traditional dyeing. Simply cationic pretreat cellulose substrates with Evertex CA-01, then proceed to salt-and alkali-free dyeing. The wastewater after dyeing is almost colorless. Process water can be easily recycled or reused.

Everclear Dyeing Solutions solve the annoying problems of reactive dyeing. In addition to saving valuable resources, it also greatly improves dyeing levelness, reproducibility, wet fastness and dyeing efficiency.

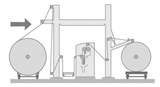
紡織基材陽離子化預處理 Cationic Pretreatment (Evertex CA-01)

針織布、毛巾布、梭織布：25℃ ~ 30℃ *16 ~ 20 小時
Knit, Towel, Woven: Cold Pad Batch 25℃ ~ 30℃ * 16 ~ 20 Hours

針織布 Knit



梭織布 Woven



後洗淨與烘乾 ; Post wash-off and drying:

a. 常溫液流洗，或是 60℃ 連續式洗淨 ;
Cold wash by Jet machine, or 60℃ continuous wash

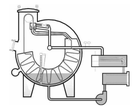
b. 洗後織物需中和到 pH6.5 ;
After wash fabric should be well neutralized to pH 6.5

c. 洗淨後織物以不超過 150℃ 連續式烘乾 ;
Post continuous drying not exceed 150℃

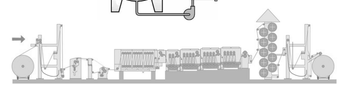
d. 接觸式烘乾不可超過 120℃ ;
not exceed 120°C when cylinder drying

染色與洗淨程序 Dyeing and Wash-off (Everclear / Everclear s/p)

針織布、毛巾布：60℃ ~ 80℃ 無鹽浸染染色＋冷水後洗淨
Knit & Towel: 60℃ ~ 80℃ salt free exhaustion dyeing
+ cold post wash-off



梭織布：連續式軋壓飽和濕蒸化染色＋ 60℃ 後洗淨
Woven: Pad Steam Dyeing + 60℃ post wash-off



傳統染色：染料 4.5%
Conventional Dyeing: Dye 4.5%

Everclear 環保染色方案：染料 2.6%
Everclear Dyeing Solution: Dye 2.6%

染色 / 後洗淨 殘餘顏色
Dyeing / Wash-off Residual



染色 / 後洗淨 殘餘顏色
Dyeing / Wash-off Residual



染色 / 後洗淨 溫度
Dyeing / Wash-off Temperature



染色 / 後洗淨 溫度
Dyeing / Wash-off Temperature



